

Rami Abou-Shamalah

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Education

MS, Applied Data Science <i>University of Michigan</i>	2023 - 2025
BS, Applied Mathematics <i>University of Western Ontario</i>	2016 - 2020

Experience

Data Scientist <i>Sander Geophysics</i>	June 2023 - Present Ottawa, ON
- Built data pipeline to fetched data from custom databases using in-house python library, conducted QA/QC and EDA to assess quality, preprocessed to remove outliers, normalize the data, and engineer features. - Developed a time series anomaly detection Machine-Learning model in ground stations recording magnetic diurnal data that decreased processing time by over 65%. - Implemented state of the art time series anomaly detection model (AnomalyTransformer) via Arxiv in PyTorch, incorporated WandB to track experiments in real-time and Hydra for configuration file management. - Developed Python programs leveraging NumPy and Pandas for statistical analysis of flight data, saving the company \$10,000 in annual savings.	
Data Analyst <i>Healthcare Systems R & A</i>	December 2021 - June 2023 Remote in Canada
- Led team of 5 in data collection and curation process: overseeing collection, digitizing, cleaning, transforming, and implementing QA/QC Standards, thus readied high quality and ML-ready training data. - Engineered geospatial features using QGIS software boosting precision by ~22% - Drove stakeholder meetings and technical presentations, breaking down domain-specifics and ML workflows for non-technical executives, ensuring strategic alignment of AI driven insights with exploration decisions	

Projects

Mineral-Prospectivity AI Platform	2025
- Designed a geospatial AI platform for mineral prospectivity using CNN-based analysis on geophysical raster datasets (gravity, magnetic). Trained deep models to identify potential hotspots for Au, Ag, Cu, Co, and Ni. - Precomputed predictions on 1 Million+ points to reduce server-side computation and stored them in Supabase. - Served user-selected regions via Supabase API, using spatial queries to dynamically return predicted results Interactive Model	
Fullstack Geospatial Data Management Platform	2024
- Built a Next.js and TailwindCSS web app with server-side rendering (SSR) - Architected a Supabase-backed system with PostGIS for geospatial queries for optimized data retrieval. - Developed a GUI for geospatial data viewing, manipulation, and real-time capabilities. - Designed and implemented relational schemas with advanced constraints and data validation using SQL, supporting diverse field collection formats (points, lines, polygons, multimedia) - Created RPC (stored procedures) to handle complex API filtering, spatial joins, permission-based access control, and optimized retrieval across large datasets (millions of rows). Live Product (still developing)	
Live Analytics Dashboard / Platform for Startup	2023
- Developed a live analytics dashboard using React, integrated it within an internal web application, enabling real-time insights into thousands of appointments across over 6 cities in a high production startup. - Enabled real-time analytics by writing Firebase Firestore listeners to fetch data when new data is added. - Secured the platform via Supabases' OAuth system, writing Firebase secret keys and authorized email secret keys within Supabases' edge function, and employing an edge function to verify if an email should have access to the data and then to pull that data, securing the platform entirely from non authorized emails. - Developed predictive analytics using time-series forecasting and seasonal decomposition based on historical data to forecast future sales in existing cities and in new cities.	